James Zhang

Senior Full-Stack AI & automation Engineer

[redacted]

Highly skilled senior Full-Stack AI & automation engineer with over 10 years of experience specializing in AI, machine learning, bot development, automation, web scraping, data processing and web development. Proven expertise in building scalable applications, deploying AI-powered solutions, and automating data pipelines, with a strong foundation in Python, JavaScript, C#, and web technologies. Adept at leveraging innovative technologies to solve complex problems and deliver impactful results.

Bachelor of Science in Computer Science

Tsinghua University, Beijing, China

Graduated: 2015

- Focused on Artificial Intelligence, Machine Learning, and Software Engineering.
- Completed coursework in Data Structures, Algorithms, Computer Vision, and Deep Learning.
- Conducted a project on image processing using OpenCV and Python, applying machine learning techniques for object detection.
- Engaged in self-driven learning of web technologies including MERN stack and cloud deployment.

FAW Group Co Ltd Changchun, China

- **Senior Full-Stack AI & Automation Engineer**

July 2023 - PRESENT

- Delivered RAG + LLM assistants for engineering docs (chunking + vector database retrieval using LLM APIs), so teams get faster, more accurate answers
- Built Python AI pipelines for computer-vision defect detection using OpenCV + TensorFlow, improving model quality via data curation and tuning
- Implemented embedding + retrieval workflows (vector DB indexing, relevance filtering) to keep generated responses accurate as content updates
- Automated data processing and analysis using Python + SQL, orchestrated with `n8n`/`Make.com`/`Zapier`, with repeatable outputs for Excel and Power BI reporting
- Built KPI dashboards and automated report generation with Power BI + Excel, including VBA macros to reduce manual formatting and consistency issues
- Shipped React/JavaScript front-end and Node.js back-end APIs that turn ML outputs into clear KPIs and operational insights
- Created scheduled web-scraping and ingestion pipelines in Python using BeautifulSoup/Selenium to keep datasets current for training and reporting
- Coordinated and mentored a team of 4 engineers (reviews, planning, CI/CD reliability) to deliver dependable production services

- **Full-Stack Developer (AI/Computer Vision)**

April 2018 - June 2023

- Delivered RAG + LLM assistants for engineering docs (chunking + vector database retrieval using LLM APIs), so teams get faster, more accurate answers
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- Junior Developer / Automation Engineer

September 2015 - March 2018

- Automated data extraction and preprocessing using Python with BeautifulSoup/Selenium, reducing manual collection effort for recurring business tasks
- Developed and maintained internal PHP + MySQL applications for task management and operational workflows
- Built ETL and data-cleaning routines in Python to standardize datasets for analysis and model evaluation
- Supported report automation and data analysis by preparing consistent outputs for Excel/BI-style consumption and validating data quality before use

- Software Engineering Intern

July 2013 - August 2015

- Assisted with ETL and data cleaning routines in Python to standardize incoming datasets for downstream processing
- Helped prototype lightweight services and scripts to transform, validate, and format data for internal tooling needs
- Contributed to front-end work with JavaScript improvements and bug fixes for internal web pages

- Supported automation tasks by documenting workflows and helping verify job outputs to ensure data pipelines ran reliably

- **AI Development**: RAG, AI Agent, AI Chatbot, LLM
- **Automation Tools:** n8n, Make.com, Zapier
- **Frameworks & Libraries:** FastAPI, Flask, MERN, Django, ASP.NET, Laravel, WordPress
- **Databases:** MySQL, PostgreSQL, SQLite, MongoDB, SQL Server, Pinecone, Qdrant
- **Tools & Platforms:** Git, Docker, AWS, Cursor, VS Code, Visual Studio
- **AI & Machine Learning Libraries:** TensorFlow, Keras, scikit-learn, OpenCV, PyTorch
- **Web Scraping:** BeautifulSoup, Scrapy, Selenium
- **Image Processing:** OpenCV, PIL/Pillow, OCR
- **Data Processing & Automation:** Data cleaning, ETL pipelines, scripting for task automation, workflow orchestration
- **Bot Development:** Building chatbots, automation bots, integration with messaging platforms (e.g., Telegram, Messenger)
- **Office Tools:** Excel, Google Docs, Google Sheet, Google Form
- **Languages:** Python, JavaScript (Node.js, React, Angular), PHP, VB.NET, C#/C++/C

- **Chinese:** Native or Bilingual Proficiency
- **English:** Full Professional Proficiency